Summary and Recommendations

Telecom Customer Churn Analysis

1. Project Overview

Customer churn is one of the major challenges faced by telecommunications companies, as losing existing customers directly affects revenue, profitability, and long-term business growth. Understanding why customers discontinue their subscriptions is essential for developing effective customer retention strategies.

This project focuses on performing Exploratory Data Analysis (EDA) on a Telecom Customer Churn dataset to identify customer behavior patterns, examine service preferences, and investigate the factors associated with customer attrition.

The dataset contains **7,043 customer records and 21 original columns**, covering customer demographics, account information, subscribed services, payment methods, billing details, and churn status.

The primary objective is to identify customer segments that demonstrate a higher tendency to discontinue services and derive actionable business insights from the available data.

2. Data Understanding and Preprocessing

The analysis began with an examination of the dataset's structure, data types, statistical summaries, missing values, and duplicate records.

The dataset contains 7,043 observations and 21 original features, including:

- **Demographic information:** Gender, Senior Citizen, Partner, and Dependents.
- **Account information:** Customer ID, Tenure, Contract, and Paperless Billing.
- **Service information:** Phone Service, Multiple Lines, Internet Service, Online Security, Online Backup, Device Protection, Tech Support, Streaming TV, and Streaming Movies.
- **Financial information:** Monthly Charges and Total Charges.
- **Target variable:** Churn.

The following preprocessing activities were performed:

1. **Data Type Conversion:** The TotalCharges column was initially stored as a string. Blank entries were replaced with zero, considering that customers with zero tenure had not accumulated any total charges. The column was subsequently converted into a numerical data type.

2. **Missing Value Inspection:** The dataset was checked for missing values using `isnull().sum()`. No missing values remained after preprocessing.
3. **Duplicate Record Verification:** Customer IDs were checked for duplicate entries. The analysis identified zero duplicate customer records.
4. **Feature Transformation:** The SeniorCitizen column, originally represented using binary values (0 and 1), was additionally converted into understandable Yes/No categories for easier interpretation.
5. **Statistical Analysis:** Descriptive statistics were examined to understand the distribution of numerical variables, including tenure, monthly charges, and total charges.

After preprocessing, the dataset contained 7,043 records with no missing values and no duplicate customer IDs.

3. Overall Customer Churn Analysis

The initial analysis examined the overall distribution of customers based on their churn status using a count plot and a percentage-based pie chart.

The findings indicate that:

- **26.54% of customers have churned**, meaning approximately one out of every four customers has discontinued the company's services.
- **73.46% of customers have remained**, indicating that the majority of the customer base continues to use the services.

This distribution highlights a significant customer retention challenge. Although the company retains most of its customers, losing more than one-quarter of its customer base represents a considerable concern.

The overall churn percentage establishes the baseline for the subsequent analysis, allowing customer segments and service categories to be examined to understand where churn is more prominent.

4. Customer Demographic Analysis

4.1 Gender-Based Churn Analysis

A count plot was created to compare customer churn across male and female customers.

The visualization shows that customer attrition occurs across both gender categories. There is no sufficiently clear indication from the count-based comparison that gender alone is a major distinguishing factor in customer churn.

This suggests that customer retention strategies should not rely exclusively on gender-based segmentation. Instead, greater attention should be given to behavioral characteristics, contract preferences, service subscriptions, and customer tenure.

4.2 Senior Citizen Analysis

The analysis examined the distribution of senior and non-senior customers and compared their churn percentages using a 100% stacked bar chart.

The findings reveal a noticeable difference between the two customer segments:

Customer Category	Retained Customers	Churned Customers
Non-Senior Citizens	76.4%	23.6%
Senior Citizens	58.3%	41.7%

Key observations:

- Among non-senior customers, 23.6% have churned, while 76.4% have remained.
- Among senior customers, 41.7% have churned, while 58.3% have remained.
- Senior citizens demonstrate an approximately 18.1 percentage-point higher churn rate than non-senior customers.
- The churn rate among senior citizens is approximately 1.77 times that of non-senior customers.

This is one of the most important findings from the demographic analysis. It indicates that senior citizens represent a comparatively higher-risk customer segment.

The company could investigate whether service complexity, customer support accessibility, pricing, or other service-related concerns contribute to this difference.

5. Customer Tenure Analysis

A histogram was created to investigate the relationship between customer tenure and churn behavior.

Tenure represents the number of months a customer has been associated with the telecommunications company.

The descriptive statistics indicate:

- Average customer tenure: **32.37 months**
- Median customer tenure: **29 months**
- Minimum tenure: 0 months

- Maximum tenure: 72 months

The histogram reveals an important behavioral pattern:

- Customers with very short tenure, particularly those who have subscribed for only one or two months, demonstrate a higher tendency to discontinue their services.
- Customers who have remained with the company for longer periods demonstrate comparatively stronger retention.
- Churn is more prominent among newer customers, indicating that the initial customer experience may play an important role in long-term retention.

This suggests that the initial stages of the customer lifecycle are particularly important.

The company should consider implementing onboarding assistance, early-stage engagement campaigns, and proactive customer support to reduce the likelihood of customers leaving shortly after subscribing.

6. Contract-Based Churn Analysis

A count plot was created to compare customer churn across three contract categories:

- Month-to-month
- One-year
- Two-year

The visualization reveals a substantial difference in churn behavior across contract types.

Key observations:

- Month-to-month contract customers account for the most noticeable churn activity.
- Customers with one-year contracts demonstrate comparatively lower churn.
- Customers with two-year contracts show the lowest churn activity among the three contract categories.

This indicates that customers with shorter contractual commitments are more likely to discontinue their subscriptions.

Longer-term contracts may encourage customer loyalty by providing greater commitment and continuity.

From a business perspective, the company could introduce attractive long-term subscription packages, loyalty benefits, and renewal incentives to encourage month-to-month customers to consider longer contracts.

7. Service-Based Churn Analysis

A comprehensive 3×3 subplot visualization was created using count plots to examine nine different service-related features.

The features analyzed include:

1. Phone Service
2. Multiple Lines
3. Internet Service
4. Online Security
5. Online Backup
6. Device Protection
7. Tech Support
8. Streaming TV
9. Streaming Movies

This visualization provides a comparative understanding of customer subscription patterns and their corresponding churn counts.

Key observations:

Phone Service and Multiple Lines

The majority of customers subscribe to Phone Service. The Multiple Lines visualization further examines how churn is distributed between customers with and without multiple-line subscriptions.

Internet Service

Fiber Optic and DSL are the two major internet service categories represented in the dataset. Customers without internet service generally demonstrate lower churn counts compared with internet-service customers.

Online Security

Customers who do not subscribe to Online Security appear to contribute a comparatively higher number of churned customers. This may indicate an opportunity to investigate the role of security-related services in customer retention.

Online Backup

Customers without Online Backup also demonstrate comparatively higher churn counts. The company could explore whether bundled backup services and subscription incentives improve customer engagement.

Device Protection

The visualization compares churn across customers with and without Device Protection, providing insights into the relationship between additional service subscriptions and customer retention.

Tech Support

Customers without Tech Support show comparatively higher churn counts than customers subscribing to this service. This highlights the potential importance of customer assistance and technical support in maintaining customer satisfaction.

Streaming TV and Streaming Movies

The two streaming-related visualizations examine customer subscription preferences and churn distribution. Their comparatively balanced patterns suggest that streaming subscriptions should be evaluated alongside other customer and service characteristics rather than considered independently.

Overall, the service analysis indicates that Online Security, Online Backup, and Tech Support are particularly interesting areas for further investigation.

However, because these visualizations primarily use customer counts, they should not be interpreted as definitive comparisons of churn rates within each service category.

8. Payment Method Analysis

A count plot was created to investigate customer churn across different payment methods.

The dataset includes four payment categories:

- Electronic Check
- Mailed Check
- Bank Transfer (Automatic)
- Credit Card (Automatic)

The visualization highlights Electronic Check as the payment category with the most noticeable churn activity.

Customers using electronic checks contribute a considerably higher number of churned customers than the other payment categories.

This observation suggests a possible relationship between payment preferences and customer attrition.

The company should investigate whether payment convenience, billing experiences, payment reliability, or customer preferences influence churn behavior.

Encouraging customers to explore automatic payment options through convenient billing arrangements and appropriate incentives could be considered as a potential retention strategy.

Importantly, the visualization presents churn counts rather than the percentage of customers churning within each payment category. Therefore, the observation should be treated as an association requiring further investigation rather than proof that electronic checks directly cause churn.

9. Key Business Insights

The overall exploratory analysis highlights several important findings:

1. **Significant Customer Attrition:** Approximately 26.54% of the total customer base has churned, indicating a need for stronger retention initiatives.
2. **Senior Citizens Represent a Higher-Risk Segment:** The churn rate among senior citizens is 41.7%, compared with 23.6% among non-senior customers.
3. **Early Customer Lifecycle Is Critical:** Customers with shorter tenure demonstrate greater churn activity, suggesting that early engagement and onboarding are important.
4. **Contract Duration Influences Retention Patterns:** Month-to-month customers demonstrate higher churn counts, while customers with longer-term contracts show stronger retention.
5. **Additional Services May Support Retention:** Customers without Online Security, Online Backup, and Tech Support contribute comparatively higher churn counts.
6. **Payment Method Requires Further Investigation:** Electronic Check users demonstrate the most noticeable churn activity in the payment-method comparison.
7. **Gender Is Not a Strong Standalone Differentiator:** Churn is observed across both gender categories, suggesting that other behavioral and account-related factors deserve greater attention.

10. Business Recommendations

Based on the findings, the following strategies could help the company improve customer retention:

1. Strengthen Early Customer Engagement

Introduce structured onboarding programs, welcome offers, and proactive follow-ups during the first few months of a customer's subscription.

2. Develop Senior Citizen-Focused Retention Strategies

Provide accessible customer support, simplified service plans, personalized assistance, and suitable subscription packages for senior customers.

3. Encourage Long-Term Contract Adoption

Introduce attractive discounts, loyalty rewards, and additional benefits for customers willing to move from month-to-month contracts to one-year or two-year subscriptions.

4. Promote Value-Added Services

Develop suitable bundles for Online Security, Online Backup, and Tech Support to increase service adoption and improve customer engagement.

5. Improve Payment Experience

Investigate the reasons behind higher churn counts among electronic check users and encourage convenient, reliable payment arrangements.

6. Implement Customer Retention Monitoring

Regularly monitor customer tenure, contract type, service subscriptions, and payment behavior to identify customers who may be at greater risk of discontinuing their subscriptions.

11. Conclusion

The Telecom Customer Churn Analysis demonstrates how exploratory data analysis and data visualization can transform raw customer information into meaningful business insights.

Through data preprocessing, statistical exploration, count plots, percentage-based visualizations, histograms, and multiple subplots, the project identifies several customer characteristics associated with churn.

The analysis particularly highlights the importance of customer tenure, contract duration, senior citizen status, additional service subscriptions, and payment methods in understanding customer attrition patterns.

The most significant quantitative finding is the difference in churn percentages between senior and non-senior customers: 41.7% versus 23.6%, respectively. Alongside the overall churn rate of 26.54%, this provides a strong starting point for developing targeted customer retention strategies.

Ultimately, the project demonstrates the value of data-driven decision-making in identifying customer retention challenges, understanding customer behavior, and supporting business strategies aimed at reducing customer attrition and improving long-term customer relationships.